

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Dibutyltin dichloride

**CAS No** 683-18-1

**Synonyms** Dibutyldichlorotin

**Product Code** 194870000; 194870010; 194870250; 194872500

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

No hazards identified

#### Health hazards

|                                                      |              |
|------------------------------------------------------|--------------|
| Acute Oral Toxicity                                  | Category 3   |
| Acute Dermal Toxicity                                | Category 4   |
| Acute Inhalation Toxicity - Dusts and Mists          | Category 2   |
| Skin Corrosion/Irritation                            | Category 1 B |
| Serious Eye Damage/Eye Irritation                    | Category 1   |
| Germ Cell Mutagenicity                               | Category 2   |
| Reproductive Toxicity                                | Category 1B  |
| Specific target organ toxicity - (repeated exposure) | Category 1   |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

#### Label Elements



Skull and Crossbones



Health Hazard



Corrosion



Environment

**Signal Word****Danger****Hazard Statements**

H301 - Toxic if swallowed  
H312 - Harmful in contact with skin  
H330 - Fatal if inhaled  
H360 - May damage fertility or the unborn child  
H314 - Causes severe skin burns and eye damage  
H341 - Suspected of causing genetic defects if inhaled  
H372 - Causes damage to organs through prolonged or repeated exposure  
H410 - Very toxic to aquatic life with long lasting effects

**Precautionary Statements**

P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P284 - Wear respiratory protection  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P405 - Store locked up  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component             | CAS No   | Weight % |
|-----------------------|----------|----------|
| Dibutyltin dichloride | 683-18-1 | <=100    |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a

|                                            |                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.                                                                                                                                                                  |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                  |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                     |
| <b>Eye Contact</b>                         | In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                               |
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                               |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                            |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                          |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Alcohol resistant foam. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Metal oxides, Hydrogen chloride gas.

### Decomposition Temperature

> 230°C

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid dust formation.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

### Methods for Containment and Clean Up

**Clean-up methods - small spillage**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe (dust, vapor, mist, gas). Avoid dust formation.

**Conditions for Safe Storage, Including any Incompatibilities**

Corrosives area. Store under an inert atmosphere. Keep container tightly closed in a dry and well-ventilated place. Keep containers tightly closed in a dry, cool and well-ventilated place. To maintain product quality: Store in freezer.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area.

Published by German Research Foundation on July 1, 2011 **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component             | Australia                                                 | New Zealand WEL | ACGIH TLV                                                         | The United Kingdom                                                            | Germany                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|-----------------------------------------------------------|-----------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dibutyltin dichloride | STEL: 0.2 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> |                 | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.2 mg/m <sup>3</sup><br>Skin | STEL: 0.2 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 0.0018 ppm (8 Stunden). AGW - exposure factor 1<br>TWA: 0.009 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>TWA: 0.004 ppm (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>Höhepunkt: 0.004 ppm<br>Höhepunkt: 0.02 mg/m <sup>3</sup> |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Neoprene       | recommendations   |                 |                 |                       |
| Natural rubber |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Particulates filter conforming to EN 143 (or AUS/NZ equivalent)

**Recommended half mask:-**

Particle filtering: EN149:2001 (or AUS/NZ equivalent)

When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties**

|                                 |                               |                                          |
|---------------------------------|-------------------------------|------------------------------------------|
| <b>Appearance</b>               | Beige                         |                                          |
| <b>Physical State</b>           | Solid                         |                                          |
| <b>Odor</b>                     | Characteristic                |                                          |
| <b>Odor Threshold</b>           | No data available             |                                          |
| <b>pH</b>                       | 2.5                           | 0.3 g/L (20°C)                           |
| <b>Melting Point/Range</b>      | 39 - 41 °C / 102.2 - 105.8 °F |                                          |
| <b>Softening Point</b>          | No data available             |                                          |
| <b>Boiling Point/Range</b>      | 135 °C / 275 °F               | @ 10 mmHg                                |
| <b>Flash Point</b>              | > 112 °C / > 233.6 °F         | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | Not applicable                | Solid                                    |
| <b>Flammability (solid,gas)</b> | No information available      |                                          |
| <b>Explosion Limits</b>         | No data available             |                                          |

|                                         |                                   |       |
|-----------------------------------------|-----------------------------------|-------|
| Vapor Pressure                          | 0.0016 mbar @ 25 °C               |       |
| Vapor Density                           | Not applicable                    | Solid |
| Specific Gravity / Density              | 1.400                             |       |
| Bulk Density                            | No data available                 |       |
| Water Solubility                        | 320 mg/L, hydrolyses in hot water |       |
| Solubility in other solvents            | No information available          |       |
| Partition Coefficient (n-octanol/water) |                                   |       |
| Component                               | log Pow                           |       |
| Dibutyltin dichloride                   | 1.56                              |       |
| Autoignition Temperature                | No data available                 |       |
| Decomposition Temperature               | > 230°C                           |       |
| Viscosity                               | Not applicable                    | Solid |
| Explosive Properties                    | No information available          |       |
| Oxidizing Properties                    | No information available          |       |

**Other information**

|                   |               |
|-------------------|---------------|
| Molecular Formula | C8 H18 Cl2 Sn |
| Molecular Weight  | 303.83        |

## Section 10 - Stability and Reactivity

|                                  |                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                                                    |
| Stability                        | Moisture sensitive.                                                                           |
| Conditions to Avoid              | Excess heat, Exposure to moist air or water.                                                  |
| Incompatible Materials           | None known.                                                                                   |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Metal oxides. Hydrogen chloride gas. |
| Hazardous Polymerization         | No information available.                                                                     |

## Section 11 - Toxicological Information

**Information on Toxicological Effects****Product Information**

|                     |            |
|---------------------|------------|
| (a) acute toxicity; |            |
| Oral                | Category 3 |
| Dermal              | Category 4 |
| Inhalation          | Category 2 |

| Component             | LD50 Oral               | LD50 Dermal | LC50 Inhalation                         |
|-----------------------|-------------------------|-------------|-----------------------------------------|
| Dibutyltin dichloride | LD50 = 50 mg/kg ( Rat ) |             | LC50 = 59 mg/m <sup>3</sup> ( Rat ) 4 h |

|                                |              |
|--------------------------------|--------------|
| (b) skin corrosion/irritation; | Category 1 B |
|--------------------------------|--------------|

|                                    |            |
|------------------------------------|------------|
| (c) serious eye damage/irritation; | Category 1 |
|------------------------------------|------------|

|                                        |                   |
|----------------------------------------|-------------------|
| (d) respiratory or skin sensitization; |                   |
| Respiratory                            | No data available |
| Skin                                   | No data available |

|                             |            |
|-----------------------------|------------|
| (e) germ cell mutagenicity; | Category 2 |
|-----------------------------|------------|

|                             |                                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| (f) carcinogenicity;        | No data available<br>There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;  | Category 1B                                                                    |
| (h) STOT-single exposure;   | No data available                                                              |
| (i) STOT-repeated exposure; | Category 1                                                                     |
| Target Organs               | Thymus.                                                                        |
| (j) aspiration hazard;      | Not applicable<br>Solid                                                        |

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## Section 12 - Ecological Information

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component             | Freshwater Fish                                  | Water Flea                                | Freshwater Algae | Microtox                                                                    |
|-----------------------|--------------------------------------------------|-------------------------------------------|------------------|-----------------------------------------------------------------------------|
| Dibutyltin dichloride | LC50: > 4 mg/L, 96h<br>semi-static (Danio rerio) | EC50: = 0.55 mg/L, 48h<br>(Daphnia magna) |                  | EC50 = 0.2 mg/L 30 min<br>EC50 = 0.33 mg/L 15 min<br>EC50 = 0.64 mg/L 5 min |

**Persistence and Degradability** Not readily biodegradable Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary

**Persistence** May persist, based on information available.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.  
**Bioaccumulative Potential** May have some potential to bioaccumulate

| Component             | log Pow | Bioconcentration factor (BCF) |
|-----------------------|---------|-------------------------------|
| Dibutyltin dichloride | 1.56    | 0.13 - 10 dimensionless       |

**Mobility** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.

## Section 14 - Transport Information

### IMDG/IMO

UN-No UN2928  
Proper Shipping Name Toxic solid, corrosive, organic, n.o.s.  
Technical Shipping Name Dibutyltin dichloride  
Hazard Class 6.1  
Subsidiary Hazard Class 8  
Packing Group II

### ADG

UN-No UN2928  
Proper Shipping Name Toxic solid, corrosive, organic, n.o.s.  
Technical Shipping Name Dibutyltin dichloride  
Hazard Class 6.1  
Subsidiary Hazard Class 8  
Packing Group II

### IATA

UN-No UN2928  
Proper Shipping Name TOXIC SOLID, CORROSIVE, ORGANIC, N.O.S.\*  
Technical Shipping Name Dibutyltin dichloride  
Hazard Class 6.1  
Subsidiary Hazard Class 8  
Packing Group II

Environmental hazards Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

Special Precautions No special precautions required

Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### Standard for the Uniform Scheduling of Medicines and Poisons

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                        | Standard for the Uniform Scheduling of Medicines and Poisons |
|----------------------------------|--------------------------------------------------------------|
| Dibutyltin dichloride - 683-18-1 | Schedule 7 listed                                            |

#### Australian Industrial Chemicals Introduction Scheme (AICIS)

| Component                        | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|----------------------------------|-------------------------------------------------------------|------------------------|
| Dibutyltin dichloride - 683-18-1 | Present                                                     | -                      |



**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component             | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-----------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Dibutyltin dichloride | X    | X     | 211-670-0 | -      | X    | X   | -    | X     | X    | X    | X     | KE-10001 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component             | CAS No   | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------------|----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Dibutyltin dichloride | 683-18-1 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

| Component             | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                   | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Dibutyltin dichloride | -                                                                   | Use restricted. See item 30. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) Use restricted. See item 20. | SVHC Candidate list - Toxic for reproduction (Article 57 c)                                           |

|  |  |                                    |  |
|--|--|------------------------------------|--|
|  |  | (see link for restriction details) |  |
|--|--|------------------------------------|--|

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Section 16 - Other Information

### Legend

**AICS** - Australian Inventory of Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical incident response training.

**Revision Date** 17-Nov-2022

**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet